

Intro Bayes

- Bayesian methods
- Bayesian models

Bayesian models

A Bayesian model attains the general form

$$q \sim Q$$
$$w|q \sim P_q$$

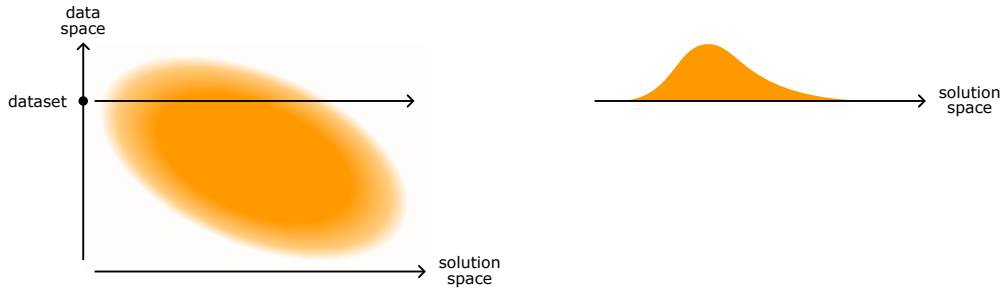
and consists of, at least, two uncertain quantities of interest

- an independent q
- a dependent w

...this is most often called the unknown parameter or solution

...this is most often called the measurement or data

Bayesian models



With a Bayesian model we seek to characterize the **posterior** probability distribution

$$p(q|w)$$

that corresponds to the **given** data

Bayesian methods

We consider a Bayesian model

$$q \sim Q$$
$$w|q \sim P_q$$

This model provides direct descriptions of:

- the **prior distribution**: this corresponds to the probability function

$$p(q) = Q(q)$$

- the **likelihood distribution**: this corresponds to the probability function

$$p(w|q) = P_q(w)$$

This model provides almost direct description of:

- the **joint distribution**: this corresponds to the probability function

$$p(w, q) = p(w|q)p(q) = P_q(w)Q(q)$$

Bayes' theorem

We consider two **parametric** quantities of interest x and y

Their probability function is

$$\begin{aligned} & p(x, y) = p(y, x) \\ \implies & p(x|y)p(y) = p(y|x)p(x) \\ \implies & p(x|y) = \frac{p(y|x)p(x)}{p(y)} \qquad p(y) \neq 0 \end{aligned}$$

This is Bayes' theorem

Bayes' rule generalizes to **any** quantity of interest, no matter if parametric or non-parametric

Bayesian methods

We consider a Bayesian model

$$q \sim Q$$
$$w|q \sim P_q$$

This model provides direct descriptions of:

- the **prior distribution**: this corresponds to the probability function

$$p(q) = Q(q)$$

- the **likelihood distribution**: this corresponds to the probability function

$$p(w|q) = P_q(w)$$

This model provides indirect description of:

- the **posterior distribution**: this corresponds to the probability function

$$p(q|w) = \frac{p(w|q)p(q)}{p(w)} \propto p(w|q)p(q) = P_q(w)Q(q)$$

Example

We consider **binomial data** with **unknown trials**

$$J \sim \text{Poisson}(\mu)$$
$$w|J \sim \text{Binomial}(J, \pi)$$

Our posterior is given by

$$\begin{aligned} p(J|w) &\propto p(w|J)p(J) \\ &= \text{Binomial}(w; J, \pi) \text{Poisson}(J; \mu) \\ &= \binom{J}{w} \pi^w (1 - \pi)^{J-w} \frac{\mu^J}{J!} e^{-\mu} \\ &= \frac{J!}{w!(J-w)!} \pi^w (1 - \pi)^{J-w} \frac{\mu^J}{J!} e^{-\mu} \\ &= \frac{\mu^J e^{-\mu}}{w!(J-w)!} \pi^w (1 - \pi)^{J-w} \\ &\propto \frac{\mu^J}{(J-w)!} (1 - \pi)^{J-w} \end{aligned}$$

We recover the missing constants with **normalization**

$$\begin{aligned} 1 &= \sum_{J=0}^{\infty} p(J|w) \\ &= \sum_{J=0}^{\infty} C \frac{\mu^J}{(J-w)!} (1 - \pi)^{J-w} \\ &= C \sum_{J=0}^{\infty} \frac{\mu^J}{(J-w)!} (1 - \pi)^{J-w} \\ &= C \mu^w e^{\mu(1-\pi)} \\ \implies p(J|w) &= \frac{\mu^{J-w}}{(J-w)!} (1 - \pi)^{J-w} e^{\mu(\pi-1)} \end{aligned}$$

Bayesian models with multiple quantities

A **Bayesian model** with M unknowns and N datapoints attains the general form

$$\begin{aligned} q^m &\sim Q^m, & m = 1, \dots, M \\ w^n | q^{1:M} &\sim P_{q^{1:M}}^n, & n = 1, \dots, N \end{aligned}$$

The priors/likelihoods may or may not differ among unknowns/datapoints

The posterior of this model is given by

$$\begin{aligned} p(q^{1:M} | w^{1:N}) &\propto p(w^{1:N} | q^{1:M}) p(q^{1:M}) \\ &= \prod_{n=1}^N p(w^n | q^{1:M}) \times \prod_{m=1}^M p(q^m) \\ &= \prod_{n=1}^N P_{q^{1:M}}^n(w^n) \times \prod_{m=1}^M Q(q^m) \end{aligned}$$

The Bayesian work-cycle

There are four main stages to probabilistic inference:

- **Likelihood:** we represent the likelihood by describing the mechanism giving rise to our data assuming hypothetical solution values
- **Prior:** we represent the prior by summarizing our beliefs about possible solution values
- **Posterior:** we derive the posterior distribution using our specific data values
- **Inference:** we use the posterior distribution to reach further conclusions as required by the task at hand

In the inference stage we may use the posterior to:

- estimate solution values ...like in parameter estimation, inverse problems, and data assimilation
- predict new data ...like in supervised learning
- cluster or classify existing data ...like in unsupervised learning
- guide the acquisition of new data ...like in experiment design, active learning, and **BO**

and many more including hypothesis testing, model comparison, and model selection